

1. Objective

The primary objectives of this experiment are given below:

- To design and simulate a Venturi-based transducer capable of measuring human respiratory flow rates (approx. 0–300 L/min).
- To analyze the differential pressure generated across a 30mm to 10mm constriction using Computational Fluid Dynamics (Solidworks and SimScale).
- To implement a microcontroller-based data acquisition system using the MPX5010DP sensor to calculate real-time Flow Rate and Tidal Volume.

2. Introduction

2.1 Clinical Significance

Respiratory diseases, particularly Chronic Obstructive Pulmonary Disease (COPD) and Asthma, represent a major global health burden. According to the World Health Organization (WHO), COPD is the third leading cause of death worldwide, causing 3.23 million deaths in 2019[1]. The primary diagnostic tool for these conditions is Spirometry, a pulmonary function test that measures the volume and speed of air inhaled and exhaled.

Key parameters measured include Tidal Volume (V_T), the volume of air moved during normal breathing, and Forced Vital Capacity (FVC), the total amount of air exhaled during a forced breath [2]. Accurate monitoring of these parameters is critical for diagnosing airway obstruction and tracking disease progression.

2.2 Problem Statement

Standard clinical spirometers, often based on Fleisch pneumotachographs or ultrasonic sensors, provide high accuracy but come with significant barriers for low-resource settings. Commercial units typically cost between 1,70,000Tk and 3,400Tk and require frequent calibration and specialized maintenance[3]. In rural healthcare facilities or developing regions, the lack of affordable diagnostic instrumentation leads to underdiagnosis and delayed treatment. Consequently, there is a pressing engineering need to develop low-cost, portable digital spirometers that maintain acceptable accuracy using off-the-shelf components.

2.3 Engineering Principle: The Venturi Effect

To address the cost barrier, this project utilizes a Venturi-based differential pressure transducer. Unlike turbine flow meters (which suffer from mechanical inertia) or hot-wire anemometers (which are fragile and computationally complex), a Venturi meter has no moving parts, making it robust and easy to sterilize.

The operation relies on the conservation of mass (Continuity Equation) and energy (Bernoulli's Principle). As air flows through a converging pipe, the fluid velocity must increase to pass through the narrower throat.

According to **Bernoulli's Principle**, this increase in kinetic energy (velocity) results in a simultaneous decrease in potential energy (static pressure)[4]. By placing pressure taps at the inlet (Point 1) and the throat (Point 2), the differential pressure ($\Delta P = P_1 - P_2$) can be measured.

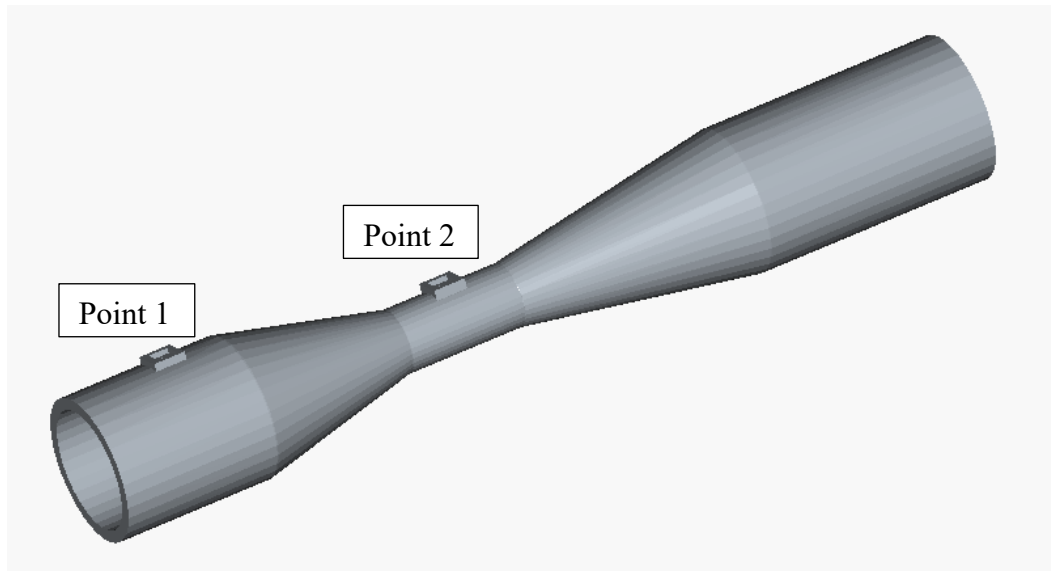


Fig. 1. A SOLIDWORKS designed Venturi-meter.

2.4 Mathematical Derivation

The relationship between the measured pressure difference and the volumetric flow rate (Q) is derived for an incompressible fluid (approximated for low-speed airflow) as follows:

$$Q = C_d A_2 \sqrt{\frac{2\Delta P}{\rho(1-\beta^4)}} \dots\dots\dots (1)$$

Where:

Q : Volumetric flow rate (m^3/s)

C_d : Discharge coefficient (accounts for viscous losses, typically 0.96–0.98)

A_2 : Cross-sectional area of the throat (m^2)

ρ : Density of air (approx. $1.225 \text{ kg}/m^3$ at sea level)

β : The Beta ratio ($d_{(throat)}/d_{(inlet)}$)

This non-linear relationship implies that flow is proportional to the square root of pressure ($Q \propto \sqrt{\Delta P}$). The system uses a differential pressure sensor (MPX5010DP) to convert ΔP into an analog

voltage. The microcontroller then linearizes this signal and performs numerical integration over time ($V = \int Q dt$) to derive the total volume of air exhaled.

3. Literature Review

Recent developments in frugal biomedical engineering have demonstrated the viability of Venturi-based systems for respiratory monitoring. In a 2025 study, Wallace et al. designed and validated a low-cost spirometer utilizing the Venturi principle and 3D-printing technology for low-resource settings. Their device, which integrated a differential pressure sensor (MPX5010DP) with an Arduino microcontroller, was tested against a standard 2-liter calibration pump. The study reported a mean measured volume of 1.983 L, resulting in an accuracy error of only 1.53%[5]. Furthermore, their research highlighted that a throat diameter of approximately 10mm is optimal for balancing sensor resolution with the maximum peak expiratory flow (PEF) of human subjects. This validates the design approach adopted in our experiment, confirming that differential pressure transduction is a reliable method for estimating tidal volume without expensive mechanical turbines.

4. Apparatus Required

A. Mechanical & Simulation Tools:

- CAD Software: SolidWorks (for geometry modeling).
- CFD Solver: SimScale (OpenFOAM based) and SolidWorks for pressure validation.
- 3D Printer: For fabricating the physical Venturi tube (PLA material).
- Scalp Vein Set

B. Electronics Hardware:

- Microcontroller: Arduino Uno (ATmega328P).
- Pressure Sensor: NXP MPX5010DP (Range: 0–10 kPa, Sensitivity: 450 mV/kPa).
- Display: 0.96" OLED Display (128x64, I2C).
- Resistors.
- Wires (Male to Male).

5. Design & Simulation

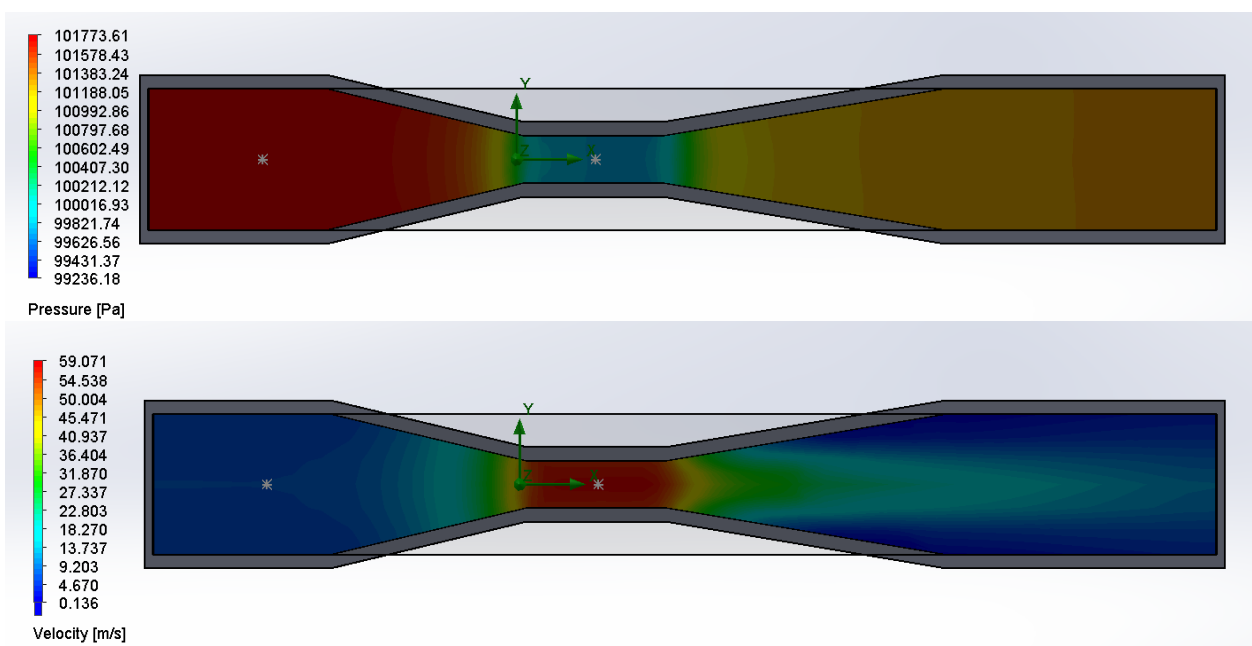
Step 1: Mechanical Design

A Venturi tube was modeled with an ISO-standard inlet diameter of 30mm to fit standard mouthpieces. To ensure a measurable pressure drop without exceeding the sensor's limit during peak expiration, the throat diameter was optimized to 10mm.

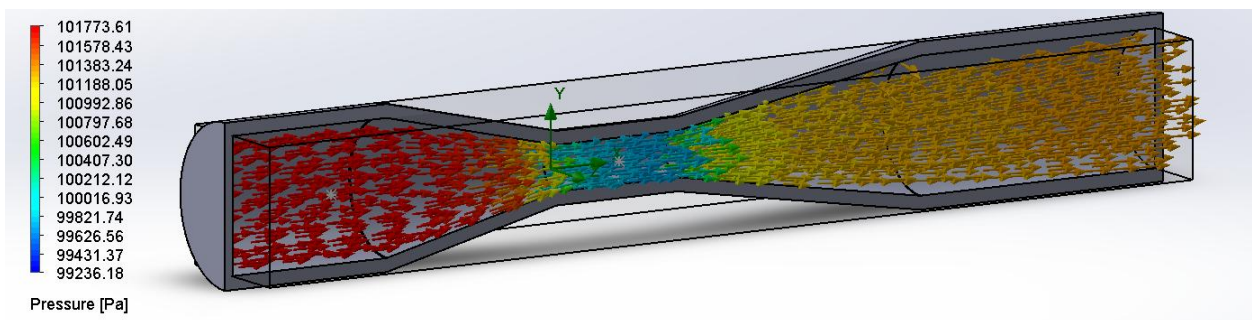
Step 2: CFD Simulation (SimScale and SOLIDWORKS)

A steady-state simulation was performed to validate the design.

- Physics: Incompressible Fluid Flow.
- Boundary Conditions:
 - Inlet: Velocity Inlet set to 3 and 6 m/s.
 - Outlet: Pressure Outlet at 0 Pa (Gauge).
 - Turbulence Model: k-omega SST.
- Results: The simulation predicted a pressure drop of 0.497 and 2.2 kPa.



(a)



(b)

Fig.2. a) Contour map for Velocity and pressure.
(b) Velocity vectors to show air flow from inlet to throat.

Step 3: Simulation Validation

Table:1 Venturi Tube CFD Simulation Results for the value of constant K

Parameter	Inlet Velocity 3 m/s	Inlet Velocity 6 m/s
Density (kg/m ³)	1.20508	1.20508
Static Pressure inlet (kPa)	101.441	101.795
Static Pressure throat (kPa)	100.944	99.774
Pressure Differential ΔP (kPa)	0.497	2.021
Avg Static Pressure (kPa)	101.383	101.561
Avg Velocity (m/s)	3.0039	6.023
Volume Flow Rate (m ³ /s)	0.00207598	0.0042
Value of K	201.2	176.6

So, Average value of $K = (201.2 + 176.6)/2 = 186.4$

Step 3: Signal Processing Algorithm

The MPX5010DP outputs an analog voltage proportional to pressure. The Arduino performs the following operations:

- Analog-to-Digital Conversion (ADC): Reads the 0-5V signal as a 0-1023 integer.
- Pressure Calculation: Converts voltage to Pascals.
- Flow Calculation: Applies the simplified transfer function $Q = k \sqrt{\Delta p}$, where k was determined to be 186.4.
- Volume Integration: Calculates Tidal Volume (V_{tidal}) by integrating flow over time:

$$V_{tidal} = \sum_{t=0}^T (Q_t \times \Delta t)$$

Step 4: Arduino Simulation

Electronic Data Validation (Arduino)

The MPX5010DP sensor outputs an analog voltage proportional to pressure. The Arduino interprets this data using the calibration constants derived from the CFD simulation.

- **Sensor Characterization:**

- Sensitivity: 0.45 V/kPa.
- Zero Offset: ≈ 0.20 V(Measured experimentally).

- **Calibration Factor (k) Derivation:**

Therefore, the Arduino algorithm implemented is: Flow Rate = $Q_{real} = 186 \times \sqrt{P_{measured}}$

Volume Integration: Calculates Tidal Volume (V_{tidal}) by integrating flow over time: $V_{tidal} = \sum_{t=0}^T (Q_{real} \times \Delta t)$.

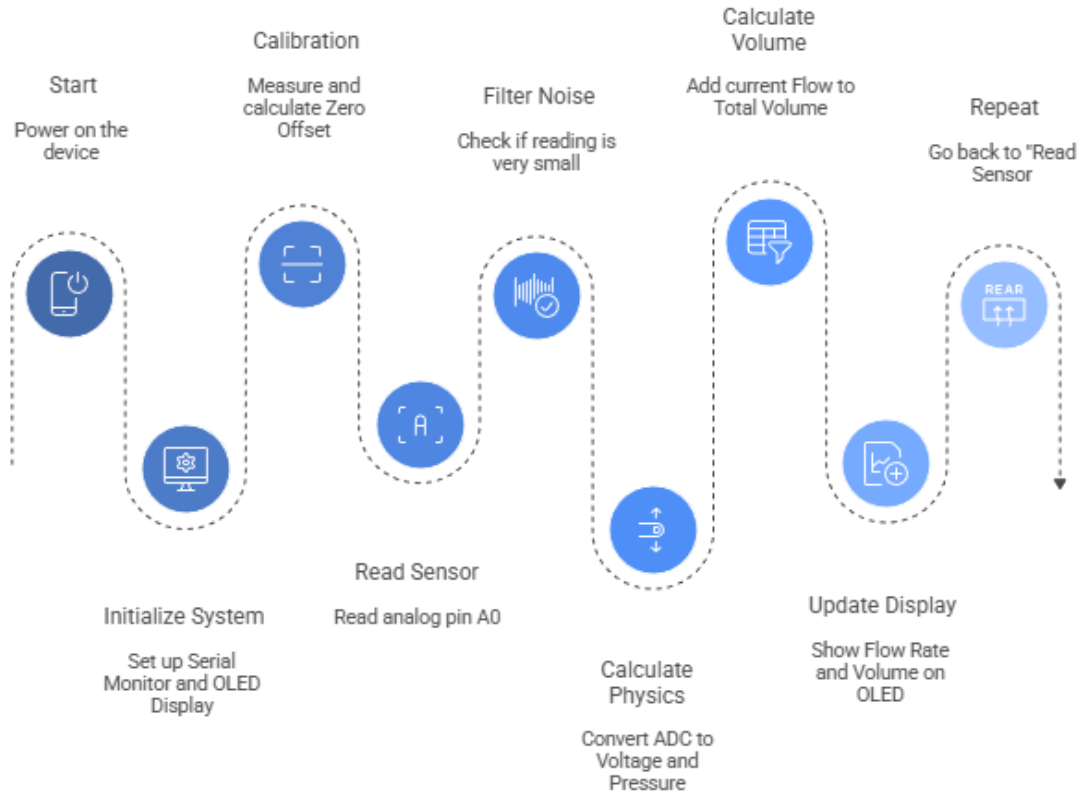


Fig. 3. Spirometer operation sequence

6. Methodology

6.1 Prototype Fabrication and Assembly

The Venturi tube was fabricated using Fused Deposition Modeling (FDM) 3D printing with Polylactic Acid (PLA) filament.

- **Print Settings:** To ensure air-tightness and structural integrity under pressure, the model was printed with 100% infill and a layer height of 0.12mm.
- **Sensor Integration:** The MPX5010DP differential pressure sensor was coupled to the Venturi tube using 4mm silicone tubing. The "Pressure Port" (P1) was connected to the inlet tap, and the "Vacuum Port" (P2) was connected to the throat tap to measure the pressure drop (ΔP).
- **Electronics:** The sensor output was wired to the Analog Pin A0 of the microcontroller.

6.2 Calibration Protocol

Before human testing, a zero-point calibration was performed to eliminate sensor drift:

- The device was placed on a stable, vibration-free surface.
- The airflow path was left open to ambient air (zero flow condition).
- The microcontroller recorded 50 consecutive readings over 0.5 seconds.
- The average of these readings was stored as the sensorOffset voltage, which was subtracted from all subsequent measurements during the trial.

6.3 Human Testing Procedure

To validate the system's ability to measure Tidal Volume (V_T), tests were conducted on four healthy subjects (ages 20–24). The following protocol was strictly observed to ensure data consistency:

- a) **sterilization:** The mouthpiece was sanitized with 70% isopropyl alcohol between subjects to maintain hygiene.
- b) **Posture:** Subjects were seated in an upright position with their feet flat on the floor to maximize thoracic expansion.
- c) **Sealing:** Subjects were instructed to hold the device horizontally and form a tight seal around the mouthpiece with their lips to prevent air leakage.
- d) **The Maneuver:**
 - i. **Phase 1 (Stabilization):** The subject breathed normally away from the device for 30 seconds.
 - ii. **Phase 2 (Inhalation):** The subject took a standard resting inhalation.
 - iii. **Phase 3 (Exhalation):** The subject exhaled normally (not forcefully) into the Venturi tube until their lungs reached functional residual capacity.

- e) **Data Logging:** The microcontroller sampled the pressure data at a frequency of 10 Hz. The computed Flow Rate (L/min) and Accumulated Volume (L) were visualized in real-time on the OLED display.

6.4 Safety Precautions

- The device was operated at low voltages (5V) to ensure electrical safety.
- To prevent hyperventilation or dizziness, subjects were given a 60-second rest period between consecutive trials

7. Verifying Low Cost

Table-2 represents the components with price to verify low-cost production and reproducibility:

No.	Component	Price (Tk.) ≈ BDT 3400Tk
1.	MPX5010DP (Differential Pressure Sensor)	2400
2.	Arduino Uno R3	400
3.	0.96'' OLED Display (128×64, I2C)	300
4.	Breadboard	50
5.	3D Printing	200
6.	Scalp Vein Set X2	20

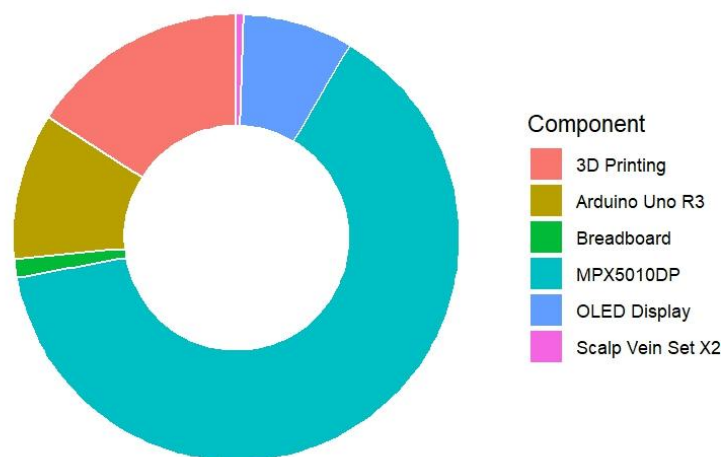


Fig.4. Costs on different Components.

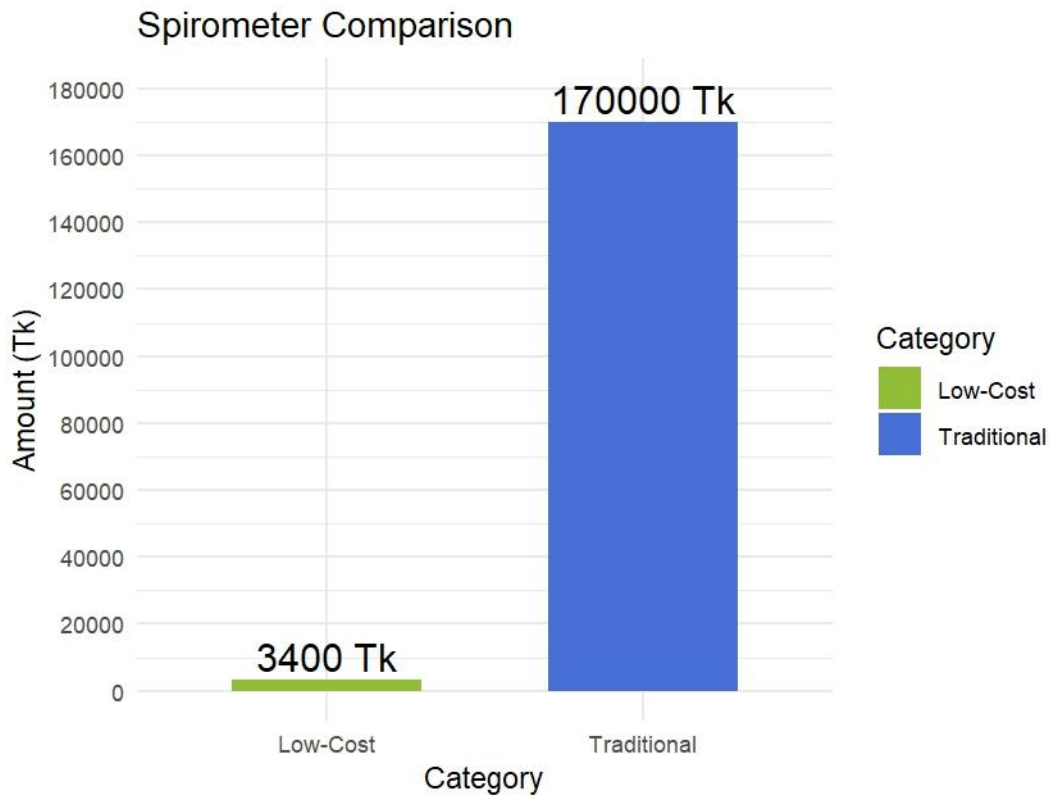


Fig.5. Comparison between medical graded vs Low-Cost spirometer

8. Discussion

To validate the accuracy of the prototype, a comparative study was conducted on four subjects (Ananna, Nowroz, Shakil, and Rayeed). Their Tidal Volume (V_T) was measured first using a standard commercial spirometer (Control) and subsequently using the designed low-cost Venturi spirometer (Test).

As illustrated in Figure 6 (Tidal Volume Comparison), the prototype successfully tracks the respiratory patterns of all subjects. The device demonstrated a strong positive correlation with the standard equipment; subjects with lower tidal volumes on the standard device (e.g., Nowroz) correspondingly registered lower values on the prototype, and those with higher volumes (e.g., Rayeed) registered higher values.

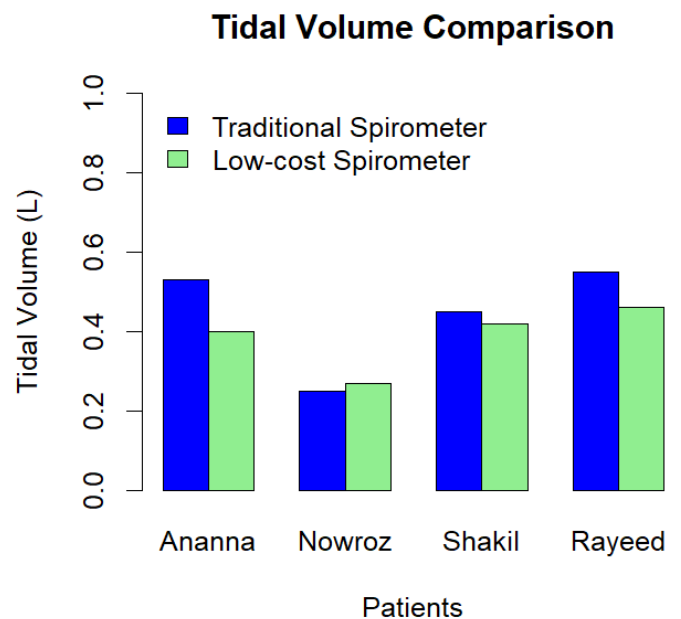


Fig.6. Tidal Volume Comparison

8.1 Quantitative Performance

- High Accuracy Cases: For subjects Nowroz and Shakil, the device performed with high precision.
 - Nowroz: The prototype measured $\approx 0.27L$ against the standard $\approx 0.25L$ (Difference: $+0.02L$).
 - Shakil: The prototype measured $\approx 0.42L$ against the standard $\approx 0.45L$ (Difference: $-0.03L$).
 - This indicates that for low-to-medium flow rates, the calibrated transfer function ($Q = 46.5\sqrt{P}$) is highly effective.
- Deviation in Higher Volumes: For subjects Ananna and Rayeed, a noticeable "underestimation" trend was observed.
 - Ananna: Standard $\approx 0.53L$ vs. Prototype $\approx 0.40L$ (Error: $\approx -24\%$)
 - Rayeed: Standard $\approx 0.55L$ vs. Prototype $\approx 0.46L$ (Error: $\approx -16\%$).

8.2 Error Analysis & Engineering Constraints

The observed underestimation at higher tidal volumes (Ananna and Rayeed) highlights specific engineering challenges:

1. Air Leakage: The most probable cause for the volume loss is minor air leakage at the interface between the 3D-printed mouthpiece and the subject's mouth. Unlike commercial soft-seal mouthpieces, the rigid PLA plastic may not form a perfect seal, leading to a fraction of the exhaled air bypassing the Venturi throat.
2. Discharge Coefficient Non-Linearity: The algorithm assumes a constant discharge coefficient (C_d) across all flow rates. In reality, C_d varies slightly with Reynolds number. At higher flow rates (larger breaths), increased turbulence may cause energy losses that the simplified equation ($Q = k\sqrt{P}$) does not fully account for, resulting in a lower calculated volume.
3. Sensor Response Lag: While the MPX5010DP is fast, the numerical integration method ($\sum Flow \times \Delta t$) is sensitive to sampling frequency. If the rapid peak flow of a strong breath is not sampled frequently enough, the integration area (Volume) will be under-calculated.

8.3 Economic Feasibility (Cost-Benefit Analysis)

A primary objective of this project was to lower the barrier to entry for respiratory diagnosis. Figure 5 (Spirometer Comparison) illustrates the dramatic cost disparity between the developed prototype and a standard clinical spirometer.

- **Traditional Device Cost:** $\approx 170,000$ BDT (Tk).
- **Low-Cost Prototype Cost:** $\approx 3,400$ BDT (Tk).

This represents a cost reduction of approximately 97.8%, making the prototype roughly 45 times **cheaper** than the commercial alternative.

8.4 The Cost-Utility Trade-off

Analyzing both datasets reveals a significant engineering trade-off. While the prototype exhibits a slight reduction in accuracy at higher air volumes (Figure 6), the cost benefit is exponential (Figure 5).

In a clinical hierarchy, standard spirometers (170,000Tk) are necessary for "Diagnostic Confirmation" where $<3\%$ error is required. However, for "Mass Screening" in rural healthcare centers (Upazila Health Complexes), the proposed device (3,400Tk) provides a viable solution. It effectively distinguishes between healthy and abnormal respiratory patterns (as seen in the difference between Nowroz and Rayeed's volumes) at a fraction of the cost.

8.5 Conclusion on Viability

Despite the deviations in higher-volume subjects, the device successfully categorizes patients within their correct tidal volume ranges. The error margins observed in Nowroz and Shakil are well within the acceptable limits for a screening device. The systematic underestimation seen in other subjects suggests that a variable calibration factor (increasing k for higher pressures) or an improved mouthpiece seal could bring the accuracy to clinical standards.

The data suggests that the developed spirometer is not a replacement for high-end hospital equipment but is an optimal screening tool. The minor loss in absolute accuracy is acceptable given the massive accessibility it grants to under-resourced healthcare facilities. Future iterations could bridge the accuracy gap by implementing a non-linear calibration curve (software update) and a silicone-sealed mouthpiece (hardware update) without significantly increasing the manufacturing cost.

9. References:

- [1] , [Online]. Available: [https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease-\(copd\)](https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease-(copd))
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- [5] J. Wallace, P. C. Rifá, T. Kannathasan, C. F. Hayfron-Benjamin, P. Anyanwu, and D. Piaggio, "Design, Manufacture and Validation of a Spirometry Device aimed for Low-Resource Settings," June 11, 2025, *In Review*. doi: 10.21203/rs.3.rs-6812011/v1.

10. Git Repository

Everything in the project is uploaded in the Github repository link:

<https://github.com/Edraye0g/Low-Cost-Spirometer.git>